Fiscal Year:	FY 2015	**Task Last Updated:**	FY 02/16/2016
PI Name:	Fitzgerald, David Jamie Ph.D.		
Project Title:	The response of articular cartilage to microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	97239-3011	**Congressional District:**	1
Comments:	NOTE: Moved to Henry Ford Health System, Detroit, MI, mid-2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	10/31/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: End date is 10/31/2014 per NSSC information (Ed., 5/22/14) NOTE: End date is now 7/31/2014 per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AQ03G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project is to determine whether the reduced mechanical forces associated with microgravity lead to osteoarthritis. Although the effects of microgravity on bone are well known and appear to be reversible, the effects reduced mechanical forces on the cartilage surfaces of articulating joints are less well understood. Studies in humans and animals clearly demonstrate that osteoarthritis results from abnormal (both increased and decreased) mechanical forces on cartilage. Based on these findings, we hypothesize that appropriate mechanical forces are critical for the health of articular cartilage. Following on from this, we further hypothesize that the absence of significant biomechanical forces affects that ability of the chondrocyte to maintain healthy functioning cartilage. This may lead to accelerated cartilage breakdown and osteoarthritis. There have been no studies that specifically investigate whether microgravity results in the degeneration of articular cartilage. Articular cartilage breakdown could be a major consequence of long-term space travel because, unlike bone, once cartilage degradation begins it cannot be reversed, typically leading to osteoarthritis. The mobility of an individual affected by even mid-stage osteoarthritis would be compromised leading to a failure to complete the full range of mission duties. The availability of joint tissue exposed to an extended period of microgravity is an ideal opportunity to address these hypotheses. Therefore, we have designed a focused Aim that will examine the joints of mice returned from microgravity for altered gene and protein expression and evidence of cartilage breakdown. This will be achieved using a range of robust and well-established methods including histology for joint surface integrity, immunohistochemistry using antibodies against a range of cartilage proteins, and gene expression analysis tools.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect to determine whether cartilage degradation occurs in space. In essence, we are looking at the changes that occur under conditions of altered biomechanical forces. The effects of increased biomechanical forces have been documented for many years but the effects of greatly reduced biomechanical forces are largely unknown. We expect that these studies will provide new insights into mechano-sensitive pathways in chondrocytes, the cells that make cartilage. This in turn may lead to the development of novel therapies and treatment options for patients with osteoarthritis and other conditions where biomechanical forces on the joint are altered.
Task Progress:	This grant supported research involving NASA's 15-day STS-131 flight and the 30-day Russian Bion-M1 mission. Mechanical forces are critical for maintaining healthy articular cartilage. Altering the mechanical forces on weight-bearing articulating joints accelerates articular cartilage degeneration leading to osteoarthritis. Conversely, the lack of mechanical forces also leads to osteoarthritis. It is clear, therefore, that normal mechanical forces are critical for the health and maintenance of articular cartilage. The long-term objective of this project is to understand the effects of microgravity on articular cartilage and specifically, whether the absence of significant mechanical forces due to microgravity results in cartilage breakdown. The presence of spaceflight-related cartilage degradation will impact multiple aspects of skeletal biology related to the failure to accommodate dynamic load, such as joint damage, bone fragility, and muscle atrophy. Summary of STS-131 findings: 15 days exposure to microgravity: For the STS-131 mission, animals were subjected to 15 days of microgravity and the joints harvested three hours following landing. There was no difference in toluidine blue staining between flight and ground articular cartilage. We conclude that 15 days exposure to microgravity does not alter proteoglycan content of articular cartilage in mouse knee joints. miRNA data: Bioinformatic analysis of the miRNA data demonstrated that 10 miRNAs were upregulated ($p>0.001$) and 26 were downregulated in flight compared to control mice. mRNA data: Thirteen mRNAs were upregulated >1.5 fold in flight compared to the ground control animals and one downregulated >1.5 fold. Of the 13 upregulated genes, 7 have been implicated in either osteoarthritis, rheumatoid arthritis, or both. Summary of Bion-M1 findings: Effects on cartilage of 30 days exposure to microgravity. There was significant proteoglycan loss, as shown by reduced toluidine staining in the knee cartilage of flight animals compared to the non-flight animals ($p=0.008$, students t-test). Interestingly, many of the chondrocytes within the middle zone remain blue. Despite this proteoglycan degradation, no clear structural changes to the articular cartilage surface of the flight mice were present suggesting that, with 30 days of unloading, the collagen II network remains intact. Conclusions Extensive literature exists in humans and in animal models suggesting that a normal range of biomechanical forces is required for cartilage homeostasis. Consistent with this is our finding that the reduced biomechanical environment of microgravity induces cartilage breakdown. Although there was no proteoglycan loss after 15 days of microgravity on STS-131, there is clear evidence of proteoglycan loss after 30 days on Bion. In addition, the mRNA changes in Bion are consistent with reduced cartilage matrix synthesis and further support the finding that cartilage atrophy occurs in spaceflight. Based on these data we speculate that the maintenance of biomechanical loading during spaceflight will reduce cartilage destruction. We believe exercise is the countermeasure. Several publications are in preparation for submission to peer-reviewed journals.
Bibliography Type:	Description: (Last Updated: 09/18/2019)

Abstracts for Journals and Proceedings	Fitzgerald J, Moscibrock C. "Articular Cartilage Degradation Following Exposure to Microgravity." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Fitzgerald J, Endicott J, Hansen U, Janowitz C. "Articular cartilage and sternal fibrocartilage respond differently to extended microgravity." npj Microgravity. 2019 Feb 18;5(1):3. https://doi.org/10.1038/s41526-019-0063-6 ; PubMed PMID: 30793021; PubMed Central PMCID: PMC6379395 , Feb-2019